

Customer Trends Data Analysis Project Summary

Project purpose

The project analyzes customer shopping behavior using a CSV dataset and applies Python, SQL, and visualization to answer business questions.

Problem statement

- Understand customer spending patterns and revenue drivers.
- Determine how discounts, subscriptions, age groups, shipping, and purchase frequency affect behavior.
- Produce insights that support dashboard reporting and business decision making.

Data source

- File: `customer\_shopping\_behavior.csv`
- Shape: 3900 rows, 18 columns
- Key fields:
 - Customer ID
 - Age
 - Gender
 - Item Purchased
 - Category
 - Purchase Amount (USD)
 - Review Rating
 - Subscription Status
 - Shipping Type
 - Discount Applied
 - Previous Purchases
 - Frequency of Purchases

Notebook workflow

The notebook loads and inspects the dataset with pandas, performs cleaning and transformation, and creates visualizations.

Data inspection

- `df.head()`
- `df.info()`
- `df.describe(include='all')`
- `df.isnull().sum()`

Data cleaning and transformation

- Filled missing `Review Rating` values using category median.
- Normalized column names to `snake\_case`.
- Renamed `purchase\_amount\_(usd)` to `purchase\_amount`.
- Added derived fields:
 - `age\_group` using age quartiles
 - `purchase\_frequency\_days` from `frequency\_of\_purchases`
- Dropped `promo\_code\_used` after verifying it matched `discount\_applied`.

Why cleaning matters

- Ensures accurate aggregation and analysis
- Prevents incorrect patterns caused by missing or inconsistent data
- Makes queries and visualizations easier to build and maintain
- Allows derived categories to answer business questions directly

Visualization

The notebook includes:

- KPI dashboard metrics
- Subscription status pie chart
- Revenue by age group bar chart

Why visualization is important

- Makes insights easier to understand quickly
- Highlights comparisons and trends
- Supports storytelling for stakeholders
- Reveals patterns not obvious from raw tables

SQL analysis file

File: `customer\_behavior\_sql\_queries.sql`

Purpose of SQL in this project

- Provides structured analytics for the relational customer table
- Supports business questions with repeatable queries
- Enables dashboard and reporting integration
- Makes it easy to compare segments and compute aggregates

Example SQL questions

- Revenue by gender
- High-spending customers using discounts
- Top products by average rating
- Compare spend across shipping types
- Subscription vs non-subscription revenue
- Discount usage by product
- Customer segmentation based on repeat purchases
- Top products per category
- Revenue by age group
- Discount impact on satisfaction and spending
- Purchase frequency effects on spending and discount usage

Why SQL instead of MongoDB / Mongoose

- The dataset is tabular and fits naturally into a relational model
- SQL is ideal for aggregation, grouping, and analytics
- PostgreSQL also uses SQL, so the approach is compatible with relational databases
- Mongoose is for MongoDB, a NoSQL document database, which is not needed here

Power BI follow-up questions

- What business story should the dashboard emphasize?
- What filters or slicers are most important?
- Which KPIs should be highlighted?
- Should the dashboard compare segments cross-sectionally or over time?

Summary

This project combines data cleaning, transformation, SQL analysis, and visualization to turn raw customer records into actionable insights. The notebook and SQL file form a strong base for a Power BI report that can communicate customer behavior across subscriptions, discounts, age groups, and product categories.